

FBISE Computer Science Solution 2nd Annual Exam, 2025

Sec A MCQs

1. Which one is a type of artificial system?

Correct option: C — Hospital Information System

A hospital information system is designed and built by people to manage information such as patient records and appointments. That makes it an **artificial system**.

- **A. Nervous system:** A natural system in a living body.
- **B. Digestive system:** Also a natural system in a living body.
- **D. Solar system:** A natural system consisting of the Sun and objects orbiting it.

Exam clue: *Artificial* means human-made.

2. Which memory type is primary and volatile?

Correct option: B — RAM

RAM is **primary memory** because the computer uses it directly while programs are running. It is **volatile** because its contents are lost when the power is turned off.

- **A. DVD:** Secondary storage; it keeps data without power.
- **C. ROM:** Primary memory, but **non-volatile**; it retains its contents without power.
- **D. Hard Disk:** Secondary storage and non-volatile.

Exam clue: Both words matter. ROM is primary, but only **RAM** is both primary *and* volatile.

3. Which unit performs arithmetic and logic operations in a computer?

Correct option: B — ALU

The **Arithmetic Logic Unit (ALU)** performs calculations, such as addition, and logical comparisons, such as checking whether one value is greater than another.

- **A. Control Unit:** Directs and coordinates operations; it does not perform the calculations.
- **C. Register:** Temporarily holds small amounts of data inside the CPU.
- **D. Cache:** Stores frequently needed data for quick access.

Exam clue: Arithmetic Logic Unit tells you its job in its name.

4. Which symbol represents processing?

Correct option: C — Rectangle

In a flowchart, a **rectangle** represents a processing step, such as calculating a total or assigning a value.

- **A. Oval:** Start or end.
- **B. Parallelogram:** Input or output.
- **D. Diamond:** A decision or condition, usually with branches such as Yes and No.

Exam clue: *Process → rectangle; decision → diamond.*

5. Main role of an algorithm in programming is:

Correct option: B — Step by step plan

An algorithm gives an ordered set of instructions for solving a problem. A programmer can then turn that plan into code.

- **A. Store data:** Data storage is a separate task.
- **C. Monitor systems:** Monitoring is not the main purpose of an algorithm.
- **D. Provide security:** Some algorithms can be used in security, but that is one application, not their general role.

Exam clue: An algorithm explains **how to solve a problem, step by step.**

6. Which HTML tag is used for the main heading?

Correct option: B — <h1>

The <h1> tag marks the main heading in the **content of a web page.**

- **A. <p>:** Marks a paragraph.
- **C. <title>:** Sets the page title shown in the browser tab; it is not the visible main heading in the page content.
- **D. <heading>:** Is not the standard HTML tag for a heading.

Exam clue: HTML headings run from <h1> to <h6>; <h1> is generally used for the main heading.

7. What is the purpose of CSS in web development?

Correct option: C — Style the web page

CSS controls how a web page looks, including its colours, fonts, spacing and layout.

- **A. Structure content:** HTML provides the page's content and structure.
- **B. Add interactivity:** JavaScript is typically used for interactive behaviour.
- **D. Store data:** Data storage is handled through storage systems or databases, not CSS.

Exam clue: HTML = content and structure; CSS = appearance; JavaScript = behavior.

8. Which JavaScript keyword declares a variable?

Correct option: A — var

var declares a variable in JavaScript. For example, var score = 10; creates a variable named score.

- **B. define:** Not a JavaScript keyword for declaring variables.
- **C. variable:** An English word, not a JavaScript declaration keyword.
- **D. input:** Not used to declare a JavaScript variable.

For similar MCQs: let and const can also declare variables in modern JavaScript, but neither appears among these options.

9. Which component transmits data between parts of the computer?

Correct option: C — Bus

A **bus** is a communication pathway that carries data and other signals between computer components, such as the CPU and memory.

- **A. Router:** Connects devices or networks; it is not the internal pathway asked about here.
- **B. Monitor:** Displays output.
- **D. Cache:** Holds frequently used data for quick access.

For similar MCQs: When the question asks how data travels *between internal components*, think **bus**.

10. Which one is an example of continuous data?

Correct option: C — Temperature

Temperature can take values throughout a range, including decimals such as 25.1°C and 25.15°C. It is **continuous data**.

- **A. Number of students:** A count; you cannot have 25.5 students.
- **B. Book titles:** Names or categories, not numerical measurements.
- **D. Postal codes:** Labels used to identify areas. Their digits do not make them continuous measurements.

For similar MCQs: Measurements such as temperature, height and time are usually continuous; counts such as the number of students are discrete.

11. How can AI be helpful in education?

Likely intended option: D — Global access

AI learning tools can help students access explanations and practice from different places. Among the given choices, **global access** is the best fit for education.

- **A. Predict earthquakes:** Not the educational use being asked about.
- **B. Calculate temperature:** Not a typical educational benefit of AI.
- **C. Play games:** Games may sometimes support learning, but merely playing games does not explain how AI helps education.

Question quality note: “Global access” is a broad benefit of online technology, not something unique to AI. A clearer answer choice would have been **personalized learning** or **instant feedback**.

12. Which one is NOT a type of malicious software?

D — Information privacy are not types of malicious software.

- **A. Malware:** Means *malicious software*. It is the broad category, rather than one specific subtype.
- **B. Spam:** Unwanted messages, often emails. It is unsolicited, unwanted bulk electronic communications (usually emails). In standard computer science curricula, spam is grouped under security threats because it is heavily used to distribute harmful links, phishing traps, and malware attachments. **Why it is incorrect here:** Spam is categorized as a cyber threat/unwanted delivery method rather than a protective privacy goal.
- **C. Spyware:** A type of malicious software that secretly collects information.
- **D. Information privacy:** The protection and appropriate handling of personal information. It is **not malicious software**.

13. Which one describes the main advantage of cloud computing?

Correct option: A — Access from anywhere

Cloud computing lets users reach files or services over the internet from different locations and suitable devices.

- **B. Increases hardware size:** Cloud computing does not make a computer’s hardware larger.
- **C. Slows processing:** That is not an advantage.
- **D. Faster internet:** Using a cloud service does not itself increase your internet connection’s speed.

Sec B Short Questions

Q.2 (i)

What is a natural system? Give two examples.

A **natural system** is a group of connected parts that exists in nature and works together without being created by humans. **Examples:** the solar system and the human digestive system.

OR — What is meant by decomposition?

Decomposition is the process of breaking a large or complex problem into smaller, manageable parts. Each part can then be understood and solved separately.

Q.2 (ii)

Write any two differences between RAM and ROM.

RAM

ROM

RAM is **volatile**: it loses its contents when power is turned off.

ROM is **non-volatile**: it retains its contents without power.

RAM temporarily stores data and programs currently in use.

ROM stores instructions or data intended to remain available, such as firmware.

OR — Write down the importance of computational thinking in problem solving.

Computational thinking helps us **break a complex problem into smaller parts** and develop a **clear, step-by-step solution**. This makes the problem easier to understand, solve, and check for errors.

Q.2 (iii)

What are input devices? Give two examples.

Input devices are hardware devices used to enter data or instructions into a computer. **Examples:** keyboard and mouse.

OR — What is the output of the given code?

The code uses `document.write()` to display text on the web page. Its intended output is:

Computer Science!

The quotation marks in the printed image appear to be typographic (“ ”). In actual JavaScript, students should type ordinary straight quotation marks, for example `document.write("Computer Science!");`.

Q.2 (iv)

What is the function of the control unit (CU) in the CPU?

The **control unit** directs and coordinates the computer’s operations. It fetches and decodes instructions, then sends control signals so the appropriate parts of the computer carry them out.

OR — Write two uses of cloud computing.

1. **Online file storage and access:** Store files in the cloud and access them from different devices.
2. **Online collaboration:** Share files so multiple people can work on them through the internet.

Q.2 (v)

Write two differences between data bus and address bus.

Data bus

Carries the **actual data** between the CPU, memory, and other components.

Usually carries data in **both directions**.

Address bus

Carries the **address** that identifies where data should be read from or written to.

Usually carries addresses **from the CPU** toward memory or an input/output device.

OR — What is the purpose of the tags **** and **<u>**?

- **** displays the enclosed text in **bold**.
- **<u>** displays the enclosed text with an **underline**.

For example, `<b>Important</b>` appears bold, while `<u>Important</u>` appears underlined.

Q.2 (vi)

Write a short note on ports. Give two examples.

A **port** is a connection point on a computer used to attach an external device and exchange data or signals with it. Two examples are a **USB port**, used for devices such as a keyboard or flash drive, and an **HDMI port**, used to send video and audio to a display.

OR — What is the purpose of information privacy?

The purpose of **information privacy** is to give people control over how their personal information is collected, used, and shared. It helps prevent unauthorized access to or misuse of that information.

Q.2 (vii)

List any two phases of design thinking.

1. **Empathize:** Understand the users and their needs.
2. **Define:** State clearly the problem that needs to be solved.

Other commonly taught phases are Ideate, Prototype, and Test.

OR — Differentiate between static and dynamic websites. Give any two points.

Static website

Usually shows the same prepared content to visitors until its files are changed.

Content is generally updated by editing the page files.

Dynamic website

Can generate or change content based on user input or stored data.

Content can often be updated through an application or database.

Q.2 (viii)

Write any two features of first-generation computers.

1. They used **vacuum tubes** as their main electronic components.

2. They were **very large and consumed a great deal of electricity**, producing considerable heat.

OR — Correct any four errors in the given HTML code.

A correctly structured version is:

```
<html>
<head>
  <title>Main Page</title>
</head>
<body>
  <p>Introduction of computer</p>
</body>
</html>
```

Corrections to identify in the exam: Close <title> with </title>; close <head> with </head>; place the paragraph text between <p> and </p>; close <body> with </body>; and close <html> with </html>. Writing the corrected code clearly shows at least four fixes.

Q.2 (ix)

Write a JavaScript statement to declare and initialize a variable.

```
let marks = 90;
```

Here, **let** declares a variable named marks, and **= 90** initializes it with the value 90.

OR — Briefly explain Von Neumann architecture.

Von Neumann architecture is a computer design in which **program instructions and data are stored in the same memory**. The CPU fetches instructions from memory and executes them, while input and output devices allow communication with the computer.

Q.2 (x)

What is the purpose of operators in JavaScript?

Operators are symbols used to perform operations on values or variables. For example, + adds numbers, while > compares two values.

OR — Write two differences between discrete and continuous data.

Discrete data

Obtained by **counting**.

Takes separate, countable values, such as **25 students**.

Continuous data

Obtained by **measuring**.

Can take any value within a range, including decimals, such as **25.5°C**.

Q.2 (xi)

Why is debugging important in programming?

Debugging is the process of finding and fixing errors in a program. It is important because it helps the program run correctly and produce the expected results.

OR — Briefly explain phishing.

Phishing is an attempt to trick people into revealing sensitive information, such as passwords or bank details. An attacker may send a message or link that **pretends to come from a trusted organization**.

Sec C Long Questions

Q.3 — Half duplex and full duplex communication

Half duplex communication allows data to travel in **both directions, but only one direction at a time**. One device sends while the other receives; they then take turns. A **walkie-talkie** is an example: one person speaks while the other listens.

Full duplex communication allows data to travel in **both directions at the same time**. Both devices can send and receive simultaneously. A **telephone call** is an example: both people can speak and hear at the same time.

Feature	Half duplex	Full duplex
Direction	Both directions	Both directions
Timing	One direction at a time	Both directions simultaneously
Example	Walkie-talkie	Telephone call

OR — Two principles of computational thinking

1. Decomposition: Breaking a complex problem into smaller, manageable parts. For example, when creating a school result program, we can separately handle entering marks, calculating totals, calculating percentages, and assigning grades.

2. Algorithm design: Writing a clear sequence of steps to solve a problem. For the same result program, an algorithm could be: enter the marks → add them → divide by total marks and multiply by 100 → apply grade rules → display the result.

Why these help: Decomposition makes a large task easier to understand, while an algorithm gives an orderly method that can be followed and checked.

Q.4 — HTML webpage with a heading, paragraph, and link

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
  <title>My Webpage</title>
```

```

</head>

<body>

    <h1>Welcome to My Webpage</h1>


    <p>This webpage provides useful learning resources.</p>


    <a href="https://google.com">Visit the Google website</a>

</body>

</html>

```

Explanation: <h1> displays the heading, <p> displays the paragraph, and <a> creates a clickable link. The href attribute specifies the link's destination. The <head> contains the browser tab title, while visible page content is placed inside <body>.

OR — System software and application software

System software manages computer hardware and provides a platform on which other programs run. **Application software** helps users perform particular tasks.

Point	System software	Application software
Main purpose	Operates and manages the computer	Performs tasks for the user
Interaction	Works mainly to support the system and applications	Used directly for activities such as writing or browsing
Examples	Windows, Linux, device drivers	Microsoft Word, Excel, web browsers

Relationship: When a user opens Word, the operating system manages resources such as memory and devices so that Word can run.

Q.5 — JavaScript program to print numbers from 10 to 1

```
for (let number = 10; number >= 1; number--) {  
    console.log(number);  
}
```

Output:

10
9
8
7
6
5
4
3
2
1

OR — Big data and three advantages

Big data refers to datasets so large, fast-growing, or varied that conventional methods may be insufficient to store and analyse them efficiently. Examples include large collections of online purchases, sensor readings, and social media activity.

Three advantages are:

1. **Better decisions:** Analysing large amounts of information can reveal useful trends and patterns.
2. **Improved services:** Organisations can understand users' needs and tailor services to them.

3. **Prediction and efficiency:** Patterns can help forecast demand and improve the use of resources.

For example, a shop can analyse sales records to predict which products it should stock.

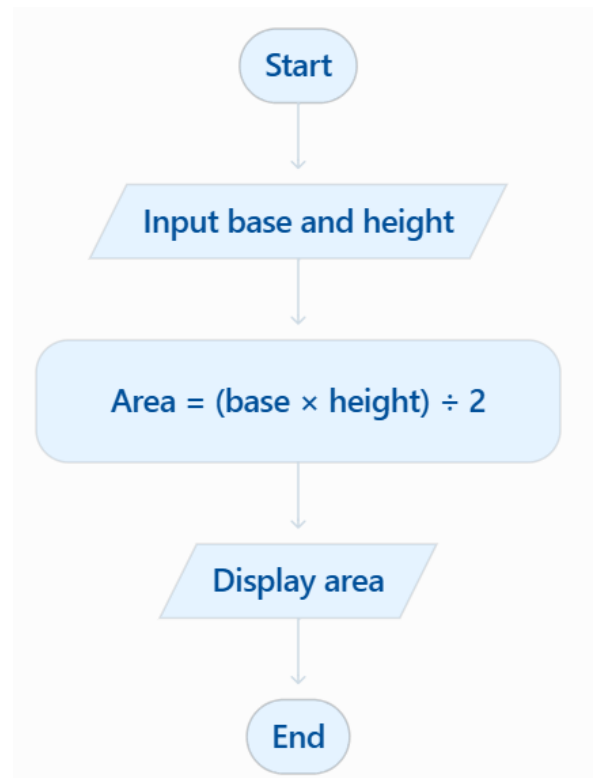
Q.6 — Artificial Intelligence and three areas

Artificial Intelligence (AI) is the field of computing concerned with creating systems that can perform tasks that normally require human intelligence, such as learning from data, understanding language, or recognising objects.

Three areas of AI are:

1. **Machine learning:** Systems learn patterns from data to make predictions or decisions. *Example:* identifying unwanted email as spam.
2. **Natural language processing (NLP):** Systems process and respond to human language. *Example:* a chatbot answering a written question.
3. **Computer vision:** Systems interpret information in images or video. *Example:* recognising road signs in a camera image.

OR — Flowchart to find the area of a triangle



FBISE Computer Science Solution Model Paper, 2025

Sec A MCQs

i. Which keyword is used to declare variables in JavaScript?

Correct option: D — const

const declares a variable whose binding cannot be reassigned:

const marks = 90;

- **A. dim:** Used in some other programming languages, but not for JavaScript variable declarations.
- **B. Var:** Incorrect capitalisation. JavaScript recognises lowercase **var**, not Var.
- **C. Let:** Incorrect capitalisation. JavaScript recognises lowercase **let**, not Let.

Exam clue: var, let and const are valid JavaScript declaration keywords, but they must be written in the correct case. Of the options *as printed*, only const is valid.

ii. Which tool is commonly used for data analysis and visualization?

Correct option: B — Microsoft Excel

Excel can organise data, perform calculations with formulas, and display results using charts.

- **A. Microsoft Word:** Mainly used to write and format documents.
- **C. Microsoft PowerPoint:** Mainly used to create presentations. It can display charts, but Excel is the better answer for analysing data.
- **D. Microsoft Outlook:** Mainly used for email, calendars and communication.

iii. Which HTML tag is used to create a hyperlink?

Correct option: D — <a>

The <a> tag creates a hyperlink. The destination is supplied with its href **attribute**:

Visit the website

- **A. <link>:** Usually connects an HTML document to an external resource, such as a CSS file; it does not create a clickable link in the page text.
- **B. <href>:** href is an **attribute**, not an HTML tag.
- **C. <hyperlink>:** Not the HTML tag used to make a hyperlink.

iv. Which statement is true about a business plan?

Correct option: A — A document outlining the goals and strategies of a business

A business plan describes what a business aims to achieve and how it plans to do so. It may also include information about customers, finances and marketing.

- **B. A list of potential customers:** This could be useful *within* a plan, but it is not the whole business plan.
- **C. A financial statement:** This reports financial information; a business plan has a wider purpose.
- **D. A marketing brochure:** This promotes a business to an audience rather than setting out its goals and strategy.

v. Which memory type is volatile?

Correct option: B — RAM

Volatile means the stored data is lost when power is switched off. RAM temporarily holds data and programs while the computer is running.

- **A. DVD:** Retains its data without power.
- **C. PROM:** Programmable read-only memory; retains its programmed contents without power.
- **D. Memory card:** Retains stored files without power.

Exam clue: If the question asks which memory *loses its contents when power goes off*, choose **RAM**.

vi. How do you display “Hello, World!” in a JavaScript alert box?

Correct option: C — `alert("Hello, World!");`

`alert()` is the JavaScript function that displays a message in an alert dialog. The message goes inside quotation marks and parentheses.

- **A. `msgBox("Hello, World!");`**; `msgBox` is not the standard JavaScript function for a browser alert.
- **B. `alertBox("Hello, World!");`**; `alertBox` is not the standard function.
- **D. `Msg("Hello, World!");`**; `Msg` is not the standard function.

Exam clue: Remember the exact function name and syntax: **`alert("message");`**

vii. Which is an example of unethical behavior online?

Correct option: B — Cyberbullying

Cyberbullying means using online messages or platforms to harass, threaten, or humiliate someone. It can harm the person targeted, so it is unethical.

- **A. Sharing positive messages:** This is generally a constructive use of social media.
- **C. Helping a friend with homework:** Helping someone learn is ethical. Doing their work dishonestly would be a different matter.
- **D. Respecting others’ privacy:** This is responsible online behavior.

Exam clue: Unethical online behavior harms others or violates their rights.

viii. What does privacy mean in computer ethics?

Correct option: B — Keeping personal information confidential and secure

Privacy means people should have control over who can access their personal information, such as passwords, phone numbers, and private messages. Others should handle that information responsibly.

- **A. Sharing personal information freely online:** This may expose information to people who should not have it.
- **C. Hacking into others’ accounts:** This violates their privacy.

- **D. Ignoring security measures:** This puts private information at risk.

Exam clue: *Privacy* concerns who is allowed to see or use personal information; *security* helps protect it.

ix. What does innovation mean in entrepreneurship?

Correct option: B — Creating something new or improving existing products or services

Innovation can mean developing a new idea **or** making an existing product, service, or way of working better. For example, a business might improve how customers order and receive its products.

- **A. Copying existing ideas:** Copying alone adds no new improvement.
- **C. Investing money in the stock market:** This is an investment activity, not the meaning of innovation.
- **D. Following others' ideas:** Following an idea without developing or improving it is not innovation.

Exam clue: Innovation involves **newness or useful improvement**.

x. Which is an example of a complex problem?

Correct option: C — Figuring out how to organize a fundraising event

Organizing an event involves several connected tasks: setting a goal, arranging a venue, planning a budget, inviting people, assigning responsibilities, and dealing with possible problems. You have to coordinate these tasks to reach the goal.

- **A. Finding the sum of two numbers:** Usually one straightforward calculation.
- **B. Memorizing a multiplication table:** A learning task, but it does not require planning and coordinating many parts.
- **D. Identifying prime numbers between 1 and 100:** This takes repeated checks, but the method and goal are well defined. Compared with the other choices, organizing the event is the clearest example of a complex problem.

Exam clue: Look for a problem with **multiple connected decisions and steps**, especially when several possible solutions exist.

xi. Which is an example of AI used in everyday life?

Correct option: D — Self-driving car

A self-driving car can use AI to interpret information from sensors, recognize road conditions, and make driving decisions.

- **A. Microwave oven:** A standard microwave follows settings or programmed controls; that alone is not AI.
- **B. Refrigerator:** A standard refrigerator regulates its temperature without needing AI.
- **C. Bicycle:** An ordinary bicycle is controlled by its rider, not an AI system.

Some modern microwaves or refrigerators may include AI features, but the question does not specify those features. **Self-driving car** is the clear answer among the options given.

xii. When making a sandwich, what is the *input*?

Correct option: C — Bread and butter

An **input** is what you start with. The bread and butter are the materials used to make the sandwich.

- **A. A finished sandwich:** This is the **output**, or result.
- **B. Spreading butter on bread:** This is the **process**, or action performed on the inputs.
- **D. Eating a sandwich:** This happens after the sandwich is made; it is not an input to making it.

Exam clue: In an input–process–output question, ask: **What do I start with?**

What do I do? What do I get? Here the answers are *bread and butter* → *spreading butter* → *finished sandwich*.

Sec B Short Questions

(i) Purpose of any three input devices

1. **Keyboard:** Enters letters, numbers, and commands into a computer.
2. **Mouse:** Moves the pointer and allows the user to select and open items.
3. **Scanner:** Converts a printed document or picture into digital form.

OR — Role of computational thinking in Computer Science

Computational thinking helps solve problems systematically by **breaking them into smaller parts**, finding patterns, and designing step-by-step instructions called **algorithms**. For example, to create a program that calculates a student's grade, we identify the inputs (marks), the process (calculate the percentage and apply grade rules), and the output (the grade).

(ii) Input, process, and output for $Y = 5x + 3$

Input Process

Output

x Multiply x by 5, then add 3: $Y = 5x + 3$ Y

For $x = -2$:

$$Y = 5(-2) + 3 = -10 + 3 = \boxed{-7}$$

Thus, **input** = -2 and **output** = -7 .

OR — HTML for the heading and ordered list

```
<h1>Software</h1>
```

```
<ol>
```

```
<li>System Software</li>
```

```
<li>Application Software</li>
```

```
</ol>
```

<h1> creates the heading, creates a numbered list, and each creates one list item.

(iii) Importance of buses and purposes of two types

Buses are communication pathways that connect computer components. They allow the CPU, memory, and other components to exchange information.

- **Data bus:** Carries the actual data between components.
- **Address bus:** Carries the address of the memory location or device to be accessed.

OR — Impact of computing on e-commerce

Computing makes it possible for businesses to **display and sell products online, process orders and digital payments, and keep records of customers and stock.** Customers can browse and place orders from different locations, while businesses can serve a wider market.

(iv) Classify the two types of data

1. **Number of students in each class — Discrete data.** It is counted and takes whole-number values, such as 25 or 26; a class cannot have 25.5 students.
2. **Heights of students in centimetres — Continuous data.** Height is measured and can include decimal values, such as 152.4 cm.

OR — Choose a storage device for a graphic designer and a backup device

The most appropriate device for **quickly saving and accessing active multimedia files is an SSD** because it has fast read and write speeds. For a **backup**, choose a **high-capacity HDD** because it can hold many large files at a lower cost per unit of storage.

A USB flash drive is portable but less suitable as the main store for a large collection. A CD/DVD has limited capacity for this purpose.

(v) Three safe and responsible search strategies

1. **Use specific search words** to find relevant results—for example, search the exact research topic instead of a single broad word.

2. **Check the reliability of sources:** Prefer recognised educational or official websites, and compare important facts across sources.
3. **Protect privacy and use information honestly:** Do not enter personal details into suspicious sites; record sources and avoid copying others' work without credit.

OR — Three suitable primary data collection methods for an eating-habits survey

1. **Questionnaire:** Ask students the same clear questions about meals and food choices; this gathers responses from many students.
2. **Interview:** Ask a smaller group follow-up questions to understand *why* they make particular food choices.
3. **Observation:** With appropriate arrangements, observe food choices in the school canteen to compare reported habits with visible choices.

Using clear, neutral questions and collecting responses from different classes makes the findings more reliable.

(vi) Output of the shown HTML code

This is a paragraph.

(Explanation: - The browser displays “**This is a paragraph.**” in red, 20 px text.

The <p> element has class="text", so the CSS rule .text applies to it. The browser tab title is “Output Question”; it does not appear in the page body.

)

OR — Three reasons wireless networks are useful

1. **Mobility:** Users can stay connected while moving within the network's coverage area.
2. **Convenience:** Devices can connect without a separate network cable for each one.

3. **Easier expansion:** New devices can usually be added without installing additional cables.

(vi) What is the output of the HTML code?

Visible output on the web page:

```
<span style="color:red;font-size:20px">This is a paragraph.</span>
```

The paragraph has class="text". The CSS rule for .text makes its **font size 20 px** and its **colour red**. “**Output Question**” appears as the browser tab title, rather than as text in the page body.

OR — Give three reasons wireless networks are useful.

1. **Mobility:** Users can stay connected while moving within the coverage area.
2. **Fewer cables:** Devices can connect without individual network cables.
3. **Easy expansion:** Additional devices can be connected without installing new cables for each one.

(vii) Why is data analytics important for determining students’ preferred ice cream flavours?

Data analytics helps the cafeteria **collect and count students’ choices, compare how popular the flavours are, and decide how much of each flavour to stock**. For example, if a survey shows that more students choose chocolate than strawberry, the cafeteria can order more chocolate and reduce unnecessary waste.

OR — Predict the JavaScript output and explain the data types.

```
let a = 10;
```

```
let b = "5";
```

```
console.log(a + b); // 105
```

```
console.log(a - b); // 5
```

Output:

105

5

a is a **number**, while b is a **string** because 5 is inside quotation marks. With +, JavaScript joins the number and string as text: "10" + "5" gives "**105**". With -, JavaScript converts the numeric string "5" to the number 5, so $10 - 5 = 5$.

(viii) Identify and correct the JavaScript errors.

The intended array contains three fruits. Write the strings with complete quotation marks and use the declared variable name **fruits**, including its final s, in both statements:

```
let fruits = ["Apple", "Banana", "Orange"];
```

```
console.log(fruits.length);
```

```
console.log(fruits.length);
```

Output:

3

3

The array has three elements, so fruits.length is **3**. The printed code uses fruit.length in the second statement; **fruit and fruits are different names** in JavaScript. Unless a separate fruit variable exists, that statement causes a ReferenceError. The quotation marks around **Apple** also need correcting as shown above.

OR — Four components of data communication

Component	Role
Sender	Device that sends the data
Message	The data being sent
Transmission medium	Path through which the data travels, such as a cable or wireless signal
Receiver	Device that receives the data

A **protocol** is also essential: it is the set of rules the sender and receiver follow to communicate. For a four-component answer, students can write the four in the table and mention the protocol as an additional point.

(ix) Compare simple and complex problems, with daily-life examples.

Simple problem

Has a clear goal and usually needs a few straightforward steps.

Example: Calculating the cost of two identical notebooks.

Complex problem

Has several connected parts and may require planning and decisions.

Example: Organising a school trip, including transport, cost, permissions, and timing.

OR — Give three benefits of AI in education.

1. **Personalised practice:** AI can suggest questions suited to a student's level.
2. **Quick feedback:** It can help students identify mistakes soon after answering.
3. **Learning support:** It can explain a concept in different ways or help students revise at their own pace.

(x) Write any three characteristics of an entrepreneur.

1. **Creative:** Finds ideas for new products or improvements.
2. **Willing to take calculated risks:** Makes decisions after considering possible gains and losses.
3. **Persistent:** Continues working through difficulties to achieve a business goal.

OR — Write three advantages of cloud computing.

1. **Access from different locations:** Files and services can be reached through the internet.
2. **Easy sharing and collaboration:** Several authorised users can work with shared files.

3. **Flexible storage and resources:** Storage or computing capacity can be increased when needed.

Sec C Long Questions

Q.3 — Natural and artificial systems

A **system** is a set of connected parts that work together for a purpose. Natural and artificial systems differ in how they come into existence.

Natural system

Exists in nature without being designed by humans.

Its parts interact through natural processes.

Example: the human digestive system.

Artificial system

Is designed and created by humans.

Its parts are arranged to perform an intended task.

Example: a hospital information system.

Natural system example: In the **digestive system**, organs such as the stomach and intestines work together to break down food and absorb nutrients.

Artificial system example: A **hospital information system** uses computers, software, and stored records to help staff manage patient information and appointments.

OR — What are AI and machine learning? How do they differ?

Artificial Intelligence (AI) is the broad field of creating computer systems that perform tasks associated with human intelligence, such as reasoning, understanding language, or recognising images.

Machine learning (ML) is a method within AI in which a computer system learns patterns from data and uses them to make predictions or decisions.

Artificial Intelligence

The **broader goal or field**: making systems perform intelligent tasks.

May use rules written by programmers or learning methods.

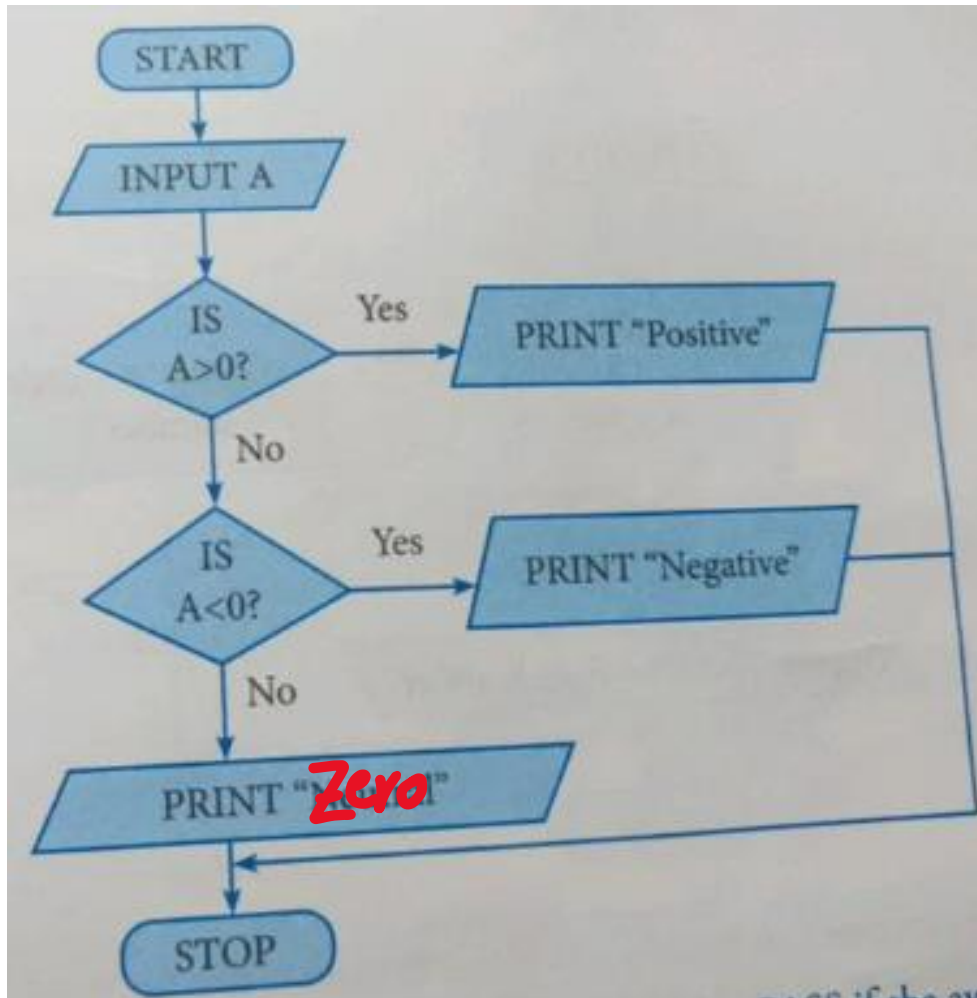
Machine Learning

A **part of AI**: learning patterns from data.

Relies on data and a learning process to improve its predictions.

Example: A system that identifies spam emails is an AI application. If it learns from examples of spam and legitimate emails, it is using **machine learning**. Thus, **ML is part of AI, but AI is broader than ML.**

Q.4 — Flowchart for positive, negative, or zero



OR — JavaScript function CalculateGrade

```
function CalculateGrade(marks) {  
    if (marks >= 90) {  
        return "A";  
    } else if (marks >= 70) {  
        return "B";  
    } else if (marks >= 50) {
```

```
    return "C";  
  } else {  
    return "F";  
  }  
}
```

Q.5 — Computing innovation and its harmful effects

A **computing innovation** is a new or improved computing product, process, or idea that uses computer technology to serve a purpose. Examples include smartphone apps, online platforms, and AI tools.

Despite their benefits, computing innovations can have harmful effects:

1. **Loss of privacy:** Apps and websites may collect personal information that can be misused or exposed in a data breach.
2. **Cybercrime and misinformation:** Digital services can be used for scams, hacking, or spreading false information.
3. **Unequal access:** People without suitable devices or internet access may be excluded from educational or economic opportunities.
4. **Electronic waste:** Discarded phones and computers can pollute the environment if they are not handled properly.
5. **Energy use:** Data centres and electronic devices consume electricity, which can contribute to environmental harm depending on how that electricity is produced.

Conclusion: Computing innovations should be designed and used responsibly, with attention to privacy, access, and environmental impact.

OR — Big data and its applications in healthcare and manufacturing

Big data refers to very large, rapidly generated, or varied collections of data that require suitable computing tools to store and analyse effectively.

Applications in healthcare:

- **Patient care:** Analysing patient records and test results can help healthcare professionals identify patterns and support treatment decisions.
- **Disease monitoring:** Analysing health data from many people can help identify disease trends and plan healthcare resources.

Applications in manufacturing:

- **Predictive maintenance:** Data from machine sensors can help identify signs of a fault before equipment breaks down.
- **Quality control:** Analysing production data can help detect defects and improve the manufacturing process.